Phone-Controlled Bluetooth Robotic Car

Project explanation

1. Objective

To build a smartphone-controlled robotic car that communicates through an HC-05 Bluetooth module and is operated from a custom mobile application. The car supports Forward, Reverse, Left, Right and Stop controls.

2. How it works

1. The phone app connects to the **HC-05** Bluetooth module.
2. Each button in the app sends one character over Bluetooth.
3. The HC-05 passes the character to the **Arduino Uno** through a software serial port.
4. The Arduino sets the input pins of the **L298N** motor driver for that command.
5. The L298N drives the two **DC motors** from the battery, so the car moves.

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Phone app --Bluetooth--> HC-05 --serial--> Arduino Uno --pins--> L298N --> DC motors
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3. Components

• Arduino Uno

• HC-05 Bluetooth module

• L298N motor driver module

• 2WD car chassis with 2 DC (BO) motors and wheels

• Battery pack, 7 to 9 V (for example 2 x 18650 cells)

• 1k and 2k resistors (voltage divider), jumper wires, switch

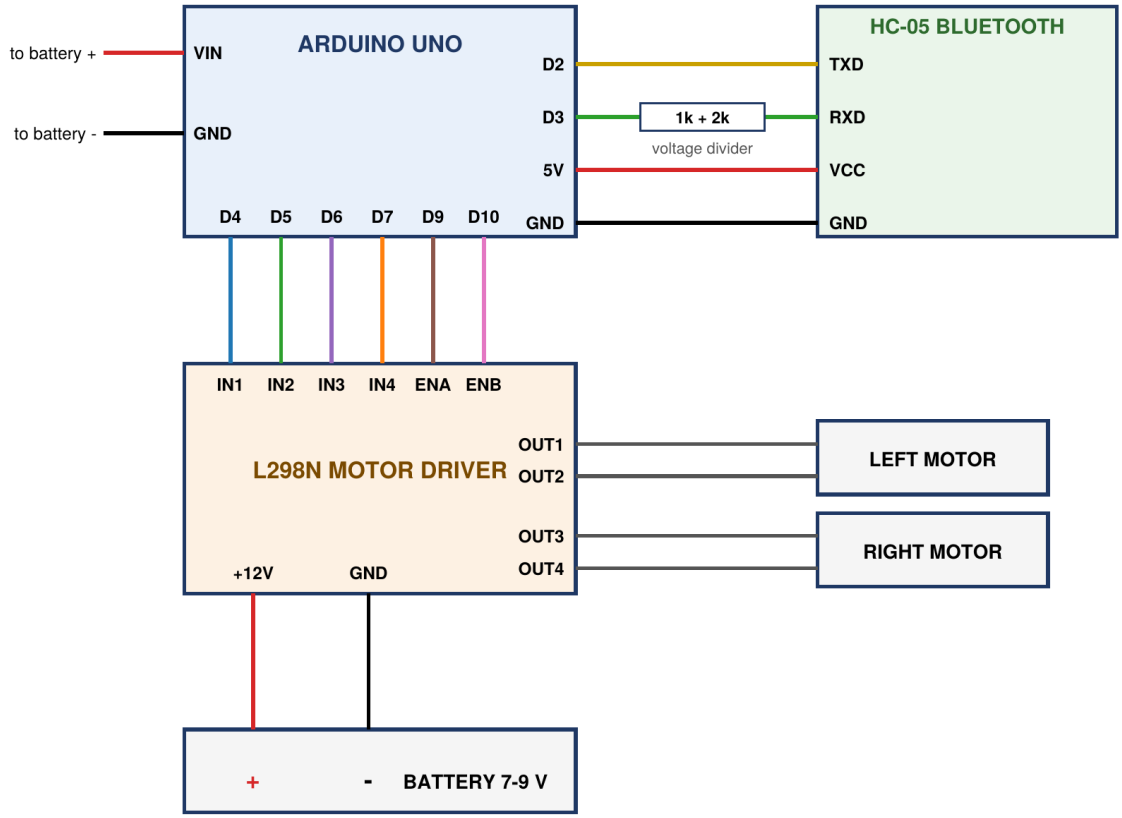
4. Command table

App button	Character sent	Car action
Forward	F	Both motors forward
Reverse	B	Both motors backward
Left	L	Left motor backward, right motor forward (turns left)
Right	R	Left motor forward, right motor backward (turns right)
Stop	S	Both motors stop
Speed slider (optional)	0 to 9	Sets motor speed (0 = slowest, 9 = fastest)

The direction buttons use press-and-hold: the app sends the direction on touch down and S on release, so the car stops when you let go.

5. Circuit and connections

Phone-Controlled Bluetooth Robotic Car - Circuit Diagram



Join all grounds together (battery -, Arduino GND, L298N GND, HC-05 GND). Battery + goes to both L298N +12V and Arduino VIN. Remove the ENA / ENB jumpers on the L298N.

From	To
HC-05 VCC	Arduino 5V
HC-05 GND	Arduino GND
HC-05 TXD	Arduino D2
HC-05 RXD	Arduino D3 through a 1k / 2k voltage divider
Arduino D4, D5	L298N IN1, IN2 (left motor)
Arduino D6, D7	L298N IN3, IN4 (right motor)
Arduino D9, D10	L298N ENA, ENB (speed, PWM)
L298N OUT1, OUT2	Left motor
L298N OUT3, OUT4	Right motor
Battery +	L298N +12V and Arduino VIN
Battery -	L298N GND and Arduino GND

• All grounds must be connected together.

• The HC-05 RXD pin accepts only 3.3 V, so the Arduino TX signal (D3) goes through a voltage divider: D3 to a 1k resistor to RXD, and a 2k resistor from RXD to GND.

• The ENA and ENB jumpers on the L298N are removed so the Arduino can control the speed.

6. Arduino program

The code (robotic_car.ino) reads one character at a time from the HC-05 at 9600 baud (the HC-05 default) using SoftwareSerial on pins D2 and D3. A switch statement calls forward(), backward(), left(), right() or stopCar(), which set the L298N direction pins. Digits 0 to 9 change the motor speed using PWM on ENA and

ENB. Unknown characters such as newlines are ignored.

7. Mobile application (MIT App Inventor)

The app has a ListPicker to choose the HC-05, a BluetoothClient, five buttons (Forward, Reverse, Left, Right, Stop) and an optional speed slider.

Event	Action
ListPicker1.BeforePicking	Set the list elements to BluetoothClient1.AddressesAndNames
ListPicker1.AfterPicking	Call BluetoothClient1.Connect with the selected address
Direction button TouchDown	SendText F, B, L or R
Direction button TouchUp	SendText S
Stop button Click	SendText S
Speed slider PositionChanged	SendText the rounded slider value

HC-05 uses classic Bluetooth, so the app works on Android phones only.

8. How to run

1. Upload robotic_car.ino to the Arduino (Board: Arduino Uno).
2. Power the car from the battery.
3. Pair the phone with HC-05 in Bluetooth settings (PIN 1234 or 0000).
4. Open the app, tap Connect to car and choose HC-05.
5. Use the buttons to drive the car.

9. Limitations

- Bluetooth range is limited to roughly 5 to 10 metres.
- Works with Android phones only (classic Bluetooth).
- There is no obstacle detection, so the car cannot avoid collisions on its own.
- Speed depends on the battery charge.